

Below is a **clear requirement draft** you can directly share with your developer team. I have aligned it with your uploaded AI/ML reference document, especially baseline engine, mode detection, anomaly detection, symptom detection, prioritization, explainability, feedback loop, edge-cloud split, and analyst workflow.

AI & ML Analysis Layer Requirement Document

✓ marks a requirement that is built and verified in the codebase as of 26 September 2026. Unmarked lines are either partly met or not started; a line is only marked when it is fully met.

For Automatic Vibration Condition Monitoring and Reliability Intelligence Platform

1. Objective

We want to develop an AI/ML-based automatic analysis layer for our vibration monitoring platform. The purpose of this layer is to help users automatically detect early-stage machine faults, understand machine health, prioritize critical equipment, generate diagnostic plots, and support analysts with clear evidence.

This AI layer should not work like a black-box system. It should provide clear reasons, plots, symptoms, fault probability, confidence level, recommended action, and historical comparison.

The system should support rotating equipment such as:

- Motors
- Pumps
- Fans
- Blowers
- Compressors
- Gearboxes
- Spindles
- Turbines
- Wind turbine drivetrain
- DG sets
- Critical industrial machines

2. Main Requirement

The AI/ML system should automatically perform the following:

1. ✓ Learn normal machine behavior.
2. Create baseline for each machine, point, axis, speed, and operating mode.
3. Detect abnormal deviation from baseline.
4. ✓ Identify possible fault symptoms.
5. ✓ Generate required plots automatically.
6. ✓ Compare current condition with past condition.
7. Detect early-stage faults before alarm limit crossing.
8. Predict machine health and reliability.
9. Estimate remaining useful life where enough trend data is available.
10. Give priority ranking of machines needing attention.
11. ✓ Generate analyst-friendly explanation.
12. Support feedback from vibration analysts to improve future analysis.

3. Asset and Equipment Details Module

The developer must create a complete equipment configuration system.

3.1 Equipment Master Data

For every machine, user should be able to enter:

- ✓ Plant name
- ✓ Area
- ✓ Line
- ✓ Machine name
- ✓ Machine ID / asset code
- ✓ Machine type
- ✓ Machine criticality
- ✓ Manufacturer
- ✓ Model
- ✓ Rated power
- ✓ Rated RPM
- ✓ Bearing details
- ✓ Coupling details
- ✓ Foundation type
- ✓ Drive type
- ✓ Load type
- ✓ Gearbox ratio if applicable
- ✓ Number of fan blades / pump vanes / gear teeth if applicable
- ✓ Motor pole count
- ✓ Bearing numbers
- ✓ Lubrication type
- ✓ Installation date
- ✓ Last maintenance date
- ✓ Process details
- ✓ Operating speed range
- ✓ Load range
- ✓ Sensor mounting location
- ✓ Sensor orientation

3.2 Measurement Point Details

For each measurement point:

- ✓ Machine side: DE / NDE / gearbox input / gearbox output / motor / pump / fan / compressor
- ✓ Direction: Horizontal / Vertical / Axial
- ✓ Sensor type: IEPE / MEMS / velocity / displacement / ultrasound / temperature
- ✓ Sensor sensitivity
- ✓ Sensor serial number
- ✓ Mounting type
- ✓ Channel number
- ✓ Sampling rate
- ✓ Frequency range
- ✓ Unit: g, mm/s, μm , gE, $^{\circ}\text{C}$, RPM

4. Settings and Configuration Module

The software should have a dedicated AI/ML settings page.

4.1 General Settings

User should be able to configure:

- ✓ Sampling rate
- ✓ FFT lines
- ✓ Frequency range
- ✓ Window type
- ✓ Averaging type
- ✓ Number of averages
- ✓ Overlap percentage
- Filter settings
- Envelope filter band
- ✓ Alarm limits
- ✓ Warning limits
- AI sensitivity level
- Baseline learning period
- Minimum data required before AI activation
- Data quality rejection rules
- RPM stability requirement
- Mode detection ON/OFF
- Auto report ON/OFF
- Auto fault detection ON/OFF

4.2 AI Sensitivity Settings

There should be selectable AI sensitivity modes:

Mode 1 — Conservative

- Fewer alarms
- Lower false positives
- Suitable for non-critical machines

Mode 2 — Balanced

- Medium sensitivity
- Recommended default mode

Mode 3 — Early Warning

- High sensitivity
- Suitable for critical machines
- Detects early-stage defects faster

Mode 4 — Expert Mode

- User can manually configure:
 - anomaly threshold
 - persistence count
 - baseline window
 - confidence threshold
 - mode separation logic
 - frequency band rules

5. Baseline Settings Module

The system must create a separate baseline for every:

- ✓ Asset
- ✓ Measurement point
- ✓ Axis
- Operating mode
- RPM band
- Load condition
- ✓ Sensor type

5.1 Baseline Learning

The system should learn normal behavior using historical healthy data.

Baseline should include:

- ✓ RMS baseline
- ✓ Peak baseline
- ✓ Crest factor baseline
- ✓ Kurtosis baseline
- ✓ FFT band energy baseline
- ✓ 1X, 2X, 3X amplitude baseline
- ✓ Bearing frequency baseline
- ✓ Envelope baseline
- ✓ Noise floor baseline
- Temperature baseline
- RPM baseline
- Load baseline

5.2 Baseline Window

Required baseline options:

- Cold start baseline: minimum 2 to 4 weeks data
- ✓ Rolling baseline: 30 to 90 days
- ✓ Manual baseline selection
- ✓ Baseline freeze option
- ✓ Baseline reset after maintenance
- Separate baseline for different operating modes

5.3 Baseline Health

System should show:

- ✓ Baseline available / not available
- ✓ Baseline confidence
- ✓ Baseline freshness
- ✓ Baseline last updated date
- ✓ Number of samples used
- Mode coverage
- ✓ Data quality of baseline

6. Operating Mode Detection

The AI should understand that one machine can operate in different conditions.

6.1 Required Modes

System should detect:

- OFF
- Idle
- Startup
- Shutdown
- Normal running
- Low load
- Medium load
- High load
- Variable speed
- Unstable operation
- Unknown mode

6.2 Inputs for Mode Detection

Mode detection should use:

- RPM
- Load
- Current
- Power
- Vibration level
- FFT shape
- Harmonic pattern
- Process signal
- Time-domain energy
- Machine state tag

6.3 Output

For each data window, system should generate:

- Mode label
- Mode confidence
- RPM band
- Stability status
- Unknown mode flag

This is very important because baseline and anomaly detection must be mode-wise. Same machine at low load and high load should not be compared blindly.

7. Automatic Plot Generation Requirement

The AI system should automatically generate plots for diagnosis.

7.1 Mandatory Plots

The system should generate:

1. ✓ Time waveform
2. ✓ Circular time waveform
3. ✓ FFT spectrum
4. ✓ Envelope spectrum
5. ✓ Trend plot
6. ✓ Waterfall plot
7. Spectrogram
8. ✓ Orbit plot where applicable

9. Bode plot where applicable
10. Polar plot where applicable
11. 1X amplitude and phase trend
12. Harmonic trend
13. Bearing frequency trend
14. Sideband trend
15. Order spectrum where RPM is available
16. Band energy trend
17. Alarm trend
18. Health score trend
19. Reliability trend
20. Remaining useful life trend

7.2 Plot Auto-Selection

AI should automatically decide which plots are useful based on suspected fault.

Example:

Bearing Fault

System should show:

- Time waveform
- Envelope spectrum
- Envelope trend
- Kurtosis trend
- BPFO/BPFI/BSF/FTF markers
- High-frequency band energy trend

Imbalance

System should show:

- FFT
- 1X trend
- Phase trend
- Bode plot
- Polar plot

Misalignment

System should show:

- FFT
- 1X, 2X, 3X harmonics
- Axial vibration trend
- Phase comparison

Looseness

System should show:

- FFT harmonic series
- Time waveform
- Subharmonic components
- Impacting pattern

Gearbox Fault

System should show:

- Gear mesh frequency

- Sidebands
 - Envelope spectrum
 - Order spectrum
 - Waterfall
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8. Feature Extraction Requirement

The AI layer must extract features from raw vibration data.

8.1 Time Domain Features

System should calculate:

- ✓ RMS
- ✓ Peak
- ✓ Peak-to-peak
- ✓ Crest factor
- ✓ Kurtosis
- ✓ Skewness
- ✓ Impulse factor
- ✓ Shape factor
- ✓ Clearance factor
- ✓ Standard deviation
- ✓ Transient burst count
- ✓ Shock index
- ✓ Modulation index
- ✓ Short-term RMS change
- ✓ Long-term RMS change

8.2 Frequency Domain Features

System should calculate:

- ✓ Dominant frequency
- ✓ 1X amplitude
- ✓ 2X amplitude
- ✓ 3X amplitude
- ✓ Harmonic count
- ✓ Harmonic energy ratio
- ✓ Sideband spacing
- ✓ Sideband energy ratio
- ✓ Spectral centroid
- ✓ Spectral spread
- ✓ Spectral entropy
- ✓ Broadband noise score
- ✓ Haystack score
- ✓ Peak drift
- ✓ Band energy
- ✓ Narrowband-to-broadband ratio

8.3 Envelope Features

System should calculate:

- ✓ BPFO energy

- ✓ BPFI energy
- ✓ BSF energy
- ✓ FTF energy
- ✓ Bearing harmonic energy
- ✓ Envelope RMS
- ✓ Envelope peak
- ✓ Envelope kurtosis
- ✓ Demodulated peak prominence
- ✓ Repetition impact frequency
- ✓ Resonance band energy

8.4 Data Quality Features

System should calculate:

- ✓ Sensor clipping
- ✓ Sensor saturation
- ✓ Noise floor
- ✓ Sensor bias drift
- Packet loss
- ✓ Missing data
- ✓ RPM instability
- ✓ Loose sensor possibility
- ✓ Abnormal DC offset
- Communication error
- ✓ Low signal trust

9. Fault Detection Requirement

The AI should detect early-stage and developed faults.

9.1 Fault Families

The system should detect:

- ✓ Unbalance
- ✓ Misalignment
- ✓ Mechanical looseness
- ✓ Bearing outer race defect
- ✓ Bearing inner race defect
- ✓ Ball / roller defect
- ✓ Cage defect
- Lubrication issue
- ✓ Gear mesh defect
- Gear tooth wear
- Gear eccentricity
- Coupling defect
- ✓ Soft foot
- Resonance
- ✓ Structural looseness
- Cavitation
- Flow turbulence

- Rotor bar defect
- ✓ Electrical looseness
- ✓ Belt defect
- ✓ Fan blade pass issue
- ✓ Pump vane pass issue
- Rub / impact
- Shaft crack indication
- Foundation weakness
- Sensor mounting issue

9.2 Fault Output

For every suspected fault, system should show:

- ✓ Fault name
- ✓ Fault family
- ✓ Severity
- ✓ Confidence
- ✓ Evidence
- Related plots
- Trend direction
- Recommended next action
- Whether immediate shutdown is needed or only inspection is required

10. Early-Stage Fault Detection

The system should not wait for ISO alarm limits only. It should detect small changes from machine-specific baseline.

10.1 Early Fault Logic

Early fault should be detected using:

- Small but persistent RMS rise
- Kurtosis rise
- Crest factor rise
- Envelope band rise
- Bearing frequency growth
- Harmonic growth
- Sideband growth
- Noise floor rise
- Modulation increase
- Repeated small impacts
- Change in FFT pattern
- Deviation from normal mode baseline

10.2 Persistence Logic

One abnormal reading should not directly become a major alarm.

System should check:

- Has the abnormality repeated?
- Is it increasing?
- Is RPM stable?
- Is sensor data trustworthy?

- Is the same symptom appearing in multiple plots?
- Is the trend accelerating?

10.3 Early Warning Output

System should create stages:

Stage 0 — Normal

No abnormal pattern.

Stage 1 — Watch

Small deviation from baseline. No action required immediately.

Stage 2 — Early Fault Suspected

Repeated abnormal pattern. Analyst review required.

Stage 3 — Developing Fault

Clear symptom growth. Inspection recommended.

Stage 4 — Severe Fault

High risk. Maintenance action required.

Stage 5 — Critical

Immediate action or shutdown recommendation.

11. Anomaly Detection Requirement

The AI should generate anomaly score for every:

- Machine
- Point
- Axis
- Feature
- Frequency band
- Event
- Operating mode

11.1 Anomaly Score

Score should be from 0 to 100.

Example:

- 0–20: Normal
- 21–40: Slight deviation
- 41–60: Watch
- 61–75: Abnormal
- 76–90: High priority
- 91–100: Critical

11.2 ML Methods

Initial practical methods:

- Robust statistical baseline
- Rolling quantile
- EWMA
- Robust Z-score
- PCA residual score
- Isolation Forest
- Clustering for mode discovery
- Rule-based symptom logic

Deep learning should not be the first version unless enough labelled data is available.

12. Health Score and Reliability Score

The system should generate clear scores.

12.1 Machine Health Score

Score from 0 to 100.

- 90–100: Excellent
- 75–89: Good
- 60–74: Watch
- 40–59: Poor
- 20–39: High risk
- 0–19: Critical

Inputs:

- Vibration severity
- Anomaly score
- Fault symptoms
- Trend acceleration
- Alarm status
- Data quality
- Machine criticality
- Maintenance history

12.2 Equipment Reliability Score

Reliability score should consider:

- Current health
- Repeated alarms
- Fault persistence
- Repair history
- Time since last maintenance
- Operating stress
- Criticality
- Trend stability
- Previous failure history

13. Remaining Useful Life Prediction

RUL should be provided only when enough historical trend data is available.

13.1 RUL Inputs

- RMS trend
- Envelope trend
- Bearing frequency trend
- Fault severity trend
- Temperature trend
- Load trend
- Operating hours
- Maintenance history
- Failure history if available

13.2 RUL Output

System should show:

- Estimated remaining useful life
- Confidence
- Degradation rate
- Trend direction
- Assumption used
- Whether RUL is reliable or not

Example:

“Estimated RUL: 45 to 70 days. Confidence: Medium. Reason: Envelope BPFO energy increased 3.2x in last 28 days and trend is accelerating.”

If data is insufficient, system should clearly show:

“RUL prediction not available due to insufficient history.”

14. AI Explanation Requirement

Every AI finding must answer these questions:

1. ✓ What changed?
2. ✓ Compared to what?
3. ✓ How strong is the evidence?
4. Which plots support this?
5. ✓ What fault is suspected?
6. How urgent is it?
7. What should the analyst do next?

Example AI Explanation

Asset: Cooling Water Pump P-204

Point: DE Horizontal

Status: High Priority

Reason:

- Envelope RMS increased 2.9x above baseline.
- Kurtosis increased by 48%.
- BPFO frequency family is visible in envelope spectrum.
- Abnormality repeated in 6 consecutive readings.
- RPM was stable.
- Sensor data quality was high.

Possible fault:

Early-stage bearing outer race defect.

Recommended action:

Review envelope spectrum, check lubrication, compare horizontal and vertical readings, and plan bearing inspection.

15. Priority and Triage Engine

The AI should automatically rank machines needing attention.

15.1 Priority Inputs

- Anomaly score
- Fault severity

- Machine criticality
- Fault persistence
- Trend acceleration
- Data quality
- Number of symptoms
- Alarm history
- Maintenance history
- Safety impact
- Production impact

15.2 Output

Priority queue should show:

- Rank
- Machine name
- Fault suspected
- Severity
- Confidence
- Recommended action
- Time since first detected
- Trend direction
- Assigned analyst
- Status

16. Analyst Feedback Loop

The system must allow analyst feedback.

16.1 Feedback Options

Analyst should be able to mark:

- Correct detection
- False alarm
- Wrong fault type
- Severity too high
- Severity too low
- Maintenance confirmed
- Fault not found
- Sensor issue
- Process-related issue
- Ignore for this machine
- Create new fault label

16.2 Why Feedback Is Required

This feedback should be stored and used for:

- Improving future AI model
 - Reducing false alarms
 - Improving prioritization
 - Improving fault classification
 - Creating site-specific intelligence
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17. Reports Requirement

AI system should generate automatic reports.

17.1 AI Report Should Include

- Asset details
- Measurement point
- Current health score
- Anomaly score
- Fault suspected
- Fault severity
- Confidence
- Baseline comparison
- Trend plots
- FFT plot
- Envelope plot
- Time waveform
- Evidence summary
- Recommended action
- Analyst comments
- Maintenance feedback
- RUL if available

17.2 Report Types

- Daily AI summary
- Weekly machine health report
- Monthly reliability report
- Critical alarm report
- Fault evolution report
- Bearing defect report
- Gearbox defect report
- Fleet health report
- Management summary report

18. Edge and Cloud Responsibility

18.1 Edge AI Responsibility

Edge should handle:

- Basic KPI calculation
- RMS, peak, kurtosis
- FFT if possible
- Data quality check
- Local threshold alarm
- Event trigger
- Local buffering
- Basic anomaly pre-screening
- Offline alert generation
- Local baseline parameter storage

18.2 Cloud AI Responsibility

Cloud should handle:

- Baseline learning
 - Baseline refresh
 - Mode clustering
 - Advanced anomaly detection
 - Fault symptom engine
 - Fleet comparison
 - RUL prediction
 - Analyst feedback learning
 - Model training
 - Model version control
 - Report generation
 - Management dashboard analytics
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19. Dashboard Requirement

19.1 Operator Dashboard

Should show:

- Machine status
- Alarm status
- Current RMS
- Current temperature
- Machine running state
- Simple action message

19.2 Analyst Dashboard

Should show:

- AI priority queue
- Waveform
- FFT
- Envelope
- Trend
- Waterfall
- Bode
- Polar
- Orbit
- Fault evidence
- Baseline comparison
- Symptom tags
- Analyst notes
- Report export

19.3 Management Dashboard

Should show:

- Top 10 risky machines
- Site health score
- Fleet reliability score
- Critical alarms

- Open findings
 - Machines under watch
 - False alarm reduction
 - Maintenance action status
 - Cost saving summary if available
-

20. AI Card Requirement

Every abnormal machine should generate one AI card.

AI Card Format

Machine:

Point:

Direction:

Current status:

AI priority:

Health score:

Anomaly score:

Fault suspected:

Confidence:

Data quality:

Baseline comparison:

Evidence:

Recommended plots:

Recommended action:

RUL estimate:

Analyst feedback status:

Example:

Machine: Pump P-101

Point: DE Horizontal

Current Status: Abnormal

AI Priority: High

Health Score: 58/100

Anomaly Score: 82/100

Fault Suspected: Early bearing outer race defect

Confidence: 0.84

Data Quality: High

Evidence:

- BPFO band energy increased 3.5x from baseline.
- Kurtosis increased 52%.
- Envelope RMS increased continuously for 18 days.
- Defect frequency visible in envelope spectrum.
- RPM stable during measurement.

Recommended Action:

Inspect bearing lubrication and plan bearing inspection during next shutdown.

21. Data Quality and Trust Score

AI should not give strong diagnosis if data quality is poor.

21.1 Data Quality Checks

System should detect:

- ✓ Sensor loose
- ✓ Sensor saturation
- ✓ Clipping
- ✓ Low signal
- ✓ High noise floor
- ✓ Missing data
- Packet loss
- Wrong RPM
- Wrong machine state
- ✓ Unstable speed
- Abnormal sensor temperature
- Communication issue

21.2 Data Trust Output

Data trust should be:

- ✓ High
- ✓ Medium
- ✓ Low
- ✓ Invalid

If trust is low, system should show:

“AI diagnosis confidence reduced due to poor signal quality.”

22. Model Governance Requirement

The system must store:

- Model version
- Baseline version
- Feature extraction version
- Processing version
- Training date
- Training data period
- Model performance
- Rollback option
- Analyst feedback history

No model should be changed silently without version tracking.

23. Acceptance Criteria

The AI/ML module will be accepted only if:

1. System can create baseline per machine, point, axis, and mode.
2. System can detect early deviation from baseline.
3. ✓ System can generate automatic FFT, waveform, envelope, and trend plots.
4. System can identify at least basic faults:
 - ✓ imbalance
 - ✓ misalignment

- ✓ looseness
 - ✓ bearing defect
 - ✓ gear defect
 - cavitation
 - rotor bar issue
5. System gives anomaly score and health score.
 6. ✓ System gives explanation for every AI finding.
 7. System shows data quality and confidence.
 8. System supports analyst feedback.
 9. System creates priority queue.
 10. System generates AI-based reports.
 11. System does not create duplicate alarms for the same repeated event.
 12. System clearly says when data is insufficient for RUL.
 13. System stores model, baseline, and processing versions.
 14. System can work with edge and cloud architecture.
 15. System supports future fleet-level intelligence.
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24. Final Development Approach

The developer should not build one single black-box AI model.

The correct architecture is modular:

1. ✓ Asset configuration module
2. ✓ Signal processing module
3. Feature extraction module
4. ✓ Baseline engine
5. Mode detection engine
6. Anomaly detection engine
7. ✓ Symptom detection engine
8. ✓ Fault diagnosis engine
9. Priority engine
10. ✓ Explainability engine
11. RUL engine
12. Report engine
13. Analyst feedback engine
14. Model governance engine

This will make the platform reliable, explainable, scalable, and suitable for real industrial condition monitoring.

Inputs Required from Domain Side for AI/ML Vibration Analysis Development

1. Machine / Asset Details

For each machine provide:

- Machine name and asset code
- Machine type: motor, pump, fan, compressor, gearbox, spindle, turbine
- Rated RPM
- Actual operating RPM range
- Power rating
- Bearing numbers
- Coupling type
- Gearbox ratio if any
- Number of blades / vanes / gear teeth if applicable
- Drive type: direct, belt, gearbox, VFD
- Machine criticality: low, medium, high, critical

2. Measurement Point Details

For every sensor point provide:

- Machine location: DE / NDE / gearbox input / output
- Direction: Horizontal / Vertical / Axial
- Sensor type
- ✓ Sensor sensitivity
- Unit: g, mm/s, gE, temperature
- Sampling rate
- Frequency range
- Channel number
- Mounting method
- Sensor serial number if available

3. Data Required

Give them sample data in this format:

- Raw time waveform
- FFT spectrum
- Envelope spectrum
- RMS trend
- Peak trend
- Kurtosis trend
- Temperature trend
- RPM trend
- Alarm history
- Maintenance history
- Failure history if available

Minimum data needed:

- Healthy machine data
- Faulty machine data
- Startup data
- Shutdown data
- Low-load data

- High-load data
- Variable-speed data

4. Baseline Rules

You need to tell them:

- How many days healthy data should be used for baseline
- Whether baseline is manual or automatic
- When baseline should freeze
- When baseline should reset after maintenance
- Separate baseline needed for different RPM/load conditions
- Minimum samples required before AI starts analysis

Recommended:

- Minimum 2–4 weeks healthy data
- Rolling baseline 30–90 days
- Separate baseline by machine, point, direction, RPM band, and load mode

5. Fault Rules from Vibration Domain

Give them basic fault logic:

Unbalance

- High 1X RPM vibration
- Mostly radial direction
- Phase stable
- Low harmonics

Misalignment

- High 1X and 2X
- Axial vibration high
- Harmonics may increase

Looseness

- Multiple harmonics of running speed
- Sometimes 0.5X / subharmonics
- Impacting in time waveform

Bearing Fault

- High kurtosis
- High crest factor
- Envelope energy rise
- BPFO / BPFI / BSF / FTF peaks visible

Gear Fault

- Gear mesh frequency
- Sidebands around gear mesh
- Harmonic growth

Cavitation

- Broadband high-frequency noise
- Random waveform
- Pump flow-related changes

Rotor Bar Fault

- 1X / electrical frequency sidebands
- 2× slip frequency sidebands around line frequency or running speed components

6. Plots Required

Tell them AI should generate these plots:

- Time waveform
- Circular time waveform
- FFT
- Envelope spectrum
- Trend plot
- Waterfall
- Spectrogram
- Bode plot
- Polar plot
- Orbit plot if displacement probes available
- Order spectrum if RPM available
- Harmonic trend
- Bearing frequency trend
- Band energy trend

7. AI Output Required

For every abnormal case, AI should output:

- Machine name
- Point name
- Direction
- Current status
- Anomaly score
- Health score
- Fault suspected
- Confidence level
- Data quality score
- Baseline comparison
- Evidence
- Recommended plots
- Recommended action
- RUL if possible
- Analyst feedback option

8. Severity Stages

Use this stage system:

- Stage 0: Normal
- Stage 1: Watch
- Stage 2: Early fault suspected
- Stage 3: Developing fault
- Stage 4: Severe fault
- Stage 5: Critical

9. Data Quality Rules

AI should check:

- ✓ Sensor loose
- ✓ Sensor clipping

- ✓ Sensor saturation
- ✓ High noise floor
- Packet loss
- ✓ Missing data
- Wrong RPM
- ✓ Unstable RPM
- Wrong operating mode
- Communication issue

If data quality is poor, AI should not give strong fault conclusion.

10. Feedback Inputs

Analyst should be able to mark:

- Correct detection
- False alarm
- Wrong fault type
- Severity too high
- Severity too low
- Sensor issue
- Maintenance confirmed
- Fault not found
- Process-related issue

11. Reports Required

Tell them system should generate:

- Daily AI summary
- Weekly machine health report
- Monthly reliability report
- Critical alarm report
- Fault evolution report
- Bearing fault report
- Gearbox report
- Fleet health report

12. Sample Output Format

Developer should make output like this:

Machine: Pump P-101

Point: DE Horizontal

Status: Early Fault Suspected

Anomaly Score: 78/100

Health Score: 62/100

Fault Suspected: Bearing outer race defect

Confidence: 0.84

Data Quality: High

Evidence:

- Envelope RMS increased 3.2x from baseline.
- Kurtosis increased by 48%.
- BPFO family visible in envelope spectrum.
- Fault repeated in 6 readings.

- RPM was stable.

Recommended Action:

Check lubrication, inspect bearing during next shutdown, and monitor daily trend.